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Comparative Analysis of Rocket Propellant Combustion Performance Using Cantera

An Extended Propellant Analysis Tool with Isp Calculation

Course: Computer Methods in Combustion (MKWS)

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Abstract

This report presents a comprehensive comparative analysis of five rocket propellant combinations — LOX/LH₂, LOX/CH₄, LOX/RP-1, N₂O₄/UDMH, and N₂O/C₂H₄ — using the Cantera thermochemical library. The developed tool extends previous work on propellant analysis by implementing a full isentropic nozzle expansion model, enabling the calculation of specific impulse (I_{sp}) previously absent from similar tools. Key parameters including adiabatic flame temperature, characteristic velocity c^* , specific impulse under vacuum and sea-level conditions, and combustion product compositions are computed across the full operational O/F ratio range for each propellant combination. Results are validated against published performance data for operational rocket engines including the SpaceX Raptor, RS-25, and Merlin 1D, with differences of 14–23%, consistent with the known overestimation of ideal equilibrium models which neglect finite-rate kinetics and boundary-layer losses. An interactive Streamlit web dashboard is provided for real-time parametric exploration. All calculations are performed at a nominal chamber pressure of 100 bar with parametric sweeps from 10 to 300 bar.

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1 Introduction

The selection of propellant combinations is one of the most critical decisions in rocket engine design, directly determining mission performance, system complexity, and safety margins. Specific impulse (I_{sp}) — the ratio of thrust to propellant mass flow rate — is the primary figure of merit, while combustion temperature, density specific impulse, and exhaust composition determine secondary constraints such as thermal management, volumetric efficiency, and environmental impact [1].

Cantera is an open-source software suite for problems involving chemical kinetics, thermodynamics, and transport processes [4]. While widely used in combustion research, its application to rocket propellant performance analysis — particularly incorporating nozzle thermodynamics for I_{sp} calculation — has been limited in academic course projects. Previous work at this institution [7] developed a Propellant Analysis Tool (PAT) in MATLAB using Cantera, calculating chamber equilibrium properties for two propellant combinations but explicitly noting the absence of I_{sp} computation as a limitation. This work directly addresses that gap.

The objectives of this project are:

1. Implement a Python-based propellant analysis tool using Cantera’s Python API with isentropic nozzle expansion for I_{sp} calculation.
2. Perform a systematic comparative analysis of five propellant combinations across operational O/F ratios and chamber pressures.
3. Validate results against published performance data of operating rocket engines.
4. Provide an interactive web-based dashboard for parametric exploration.

2 Literature Review

2.1 Rocket propellant thermodynamics

The theoretical performance of a rocket propellant is determined by the equilibrium state of the combustion products in the chamber and the subsequent isentropic expansion through the nozzle. The adiabatic flame temperature T_{ad} for a given propellant mixture is found by solving the enthalpy balance at constant pressure [2]:

$$H_{reactants}(T_0, p_c) = H_{products}(T_{ad}, p_c) \quad (1)$$

Cantera solves this problem using the `equilibrate('HP')` command, which finds the chemical equilibrium composition and temperature at constant enthalpy and pressure

[5].

2.2 Specific impulse and nozzle performance

The specific impulse is defined as:

$$I_{sp} = \frac{V_e}{g_0} \quad (2)$$

where V_e is the effective exhaust velocity and $g_0 = 9.806\,65 \text{ m/s}^2$ is the standard gravitational acceleration. For an ideal nozzle with isentropic expansion, the exit velocity can be determined from the enthalpy drop between the chamber state and the exit state:

$$V_e = \sqrt{2(h_c - h_e)} \quad (3)$$

where h_c is the specific enthalpy at chamber conditions and h_e is the specific enthalpy at the nozzle exit, obtained after isentropic expansion to the exit pressure p_e . The characteristic velocity is given by:

$$c^* = \frac{\sqrt{\gamma RT_c}}{\Gamma} \quad (4)$$

where $\Gamma = \sqrt{\gamma} \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/[2(\gamma-1)]}$ is the Vandekerckhove function [3].

2.3 Propellant combinations under study

The five propellant combinations analysed in this work span the range of practical rocket propellants:

- **LOX/LH₂** — highest specific impulse, used in the Space Shuttle Main Engine (SSME/RS-25) and Ariane 5 Vulcain engine. Storage requires cryogenic temperatures for both components [3].
- **LOX/CH₄** — chosen for the SpaceX Raptor engine due to its advantages in propellant production on Mars and compatibility with full-flow staged combustion cycles [8].
- **LOX/RP-1** — kerosene-based propellant used in SpaceX Merlin, RD-180, and the historic F-1 engine of Saturn V [1].
- **N₂O₄/UDMH** — storable hypergolic propellant used in the Soviet/Russian Proton launch vehicle [9].
- **N₂O/C₂H₄** — self-pressurising, commonly used in student and amateur hybrid rocket projects [10].

3 Model Description

3.1 Thermochemical framework

All thermochemical calculations are performed using Cantera 3.x with the GRI-Mech 3.0 reaction mechanism (`gri30.yaml`), which comprises 325 elementary reactions among 53 species [6]. The mechanism was chosen for its comprehensive coverage of hydrogen, methane, ethane, ethylene, and nitrogen oxide chemistry.

3.2 Chamber equilibrium calculation

For each propellant combination at a given O/F ratio and chamber pressure p_c , the chamber state is determined by:

1. Initialising a Cantera `Solution` object with `gri30.yaml`.
2. Setting the mixture composition according to the O/F mass ratio:

$$x_{ox} = \frac{O/F \cdot M_{fuel}}{M_{ox}} \quad (5)$$

where M_{fuel} and M_{ox} are the molar masses of fuel and oxidizer.

3. Calling `gas.equilibrate('HP')` to find the adiabatic equilibrium state at constant enthalpy and pressure.

The following chamber parameters are extracted: T_c , $\gamma = c_p/c_v$, $\bar{M}$ (mean molecular weight), c^* , sound speed, and mole fractions of the ten most abundant species.

3.3 Nozzle exit conditions and I_{sp} calculation

The isentropic nozzle expansion is modelled by:

1. Recording the chamber enthalpy h_c and entropy s_c after equilibration.
2. Setting the gas state to (s_c, p_e) — the same entropy at exit pressure.
3. Calling `gas.equilibrate('SP')` to obtain the shifted equilibrium composition at exit conditions.
4. Computing the exit velocity from equation (3) and specific impulse from equation (2).

Two exit pressure conditions are considered: $p_{e,vac} = 0.1$ Pa (vacuum, representative of upper-stage operation) and $p_{e,SL} = 101\,325$ Pa (sea level, representative of first-stage operation).

3.4 Parametric study

The following parameter space is explored:

- O/F ratio: specific range for each propellant (see Table 1)
- Chamber pressure: $p_c \in \{10, 30, 50, 100, 200, 300\}$ bar
- Initial temperature: $T_0 = 300$ K for all cases

Table 1: O/F ratio sweep ranges and stoichiometric values for each propellant.

Propellant	Fuel	O/F min	O/F max	Stoich. O/F
LOX/LH ₂	H ₂	2.0	10.0	7.94
LOX/CH ₄	CH ₄	1.5	6.0	3.97
LOX/C ₂ H ₅ OH	C ₂ H ₅ OH	1.0	4.0	2.09
N ₂ O ₄ /UDMH	N ₂ H ₄	0.5	3.0	2.07
N ₂ O/C ₂ H ₄	C ₂ H ₄	2.0	8.0	6.67

[†] C₂H₅OH substituted by C₂H₆ (ethane); N₂H₄ substituted by NH₃ (ammonia) — species unavailable in gri30.yaml.

3.5 Software implementation

The tool is implemented in Python 3.11 and consists of three modules:

- `propellant_analysis.py` — core Cantera calculation engine
- `plots.py` — publication-quality figure generation (matplotlib)
- `app.py` — interactive Streamlit web dashboard

A summary of the key function implementing the I_{sp} calculation is given below:

```

1 import cantera as ct
2 import numpy as np
3
4 G0 = 9.80665 # standard gravity [m/s^2]
5
6 def calculate_isp(gas, p_exit):
7     """
8     Compute Isp via isentropic expansion to exit pressure.
9
10    Parameters
11    -----
12    gas : ct.Solution
13        Cantera gas object after equilibrate('HP') in chamber.
14    p_exit : float
15        Nozzle exit pressure [Pa].
16

```

```

17     Returns
18     -----
19     isp : float
20         Specific impulse [s].
21     """
22     h_chamber = gas.enthalpy_mass      # chamber total enthalpy [J/
kg]
23     s_chamber = gas.entropy_mass      # chamber entropy [J/kg/K]
24
25     # Isentropic expansion to exit pressure
26     gas.SP = s_chamber, p_exit
27     gas.equilibrate('SP')
28
29     h_exit = gas.enthalpy_mass         # exit enthalpy [J/kg]
30     delta_h = h_chamber - h_exit      # enthalpy drop [J/kg]
31
32     if delta_h <= 0:
33         return 0.0                    # non-physical, skip
34
35     V_exit = np.sqrt(2.0 * delta_h)   # exit velocity [m/s]
36     return V_exit / G0                # specific impulse [s]

```

Listing 1: Core function for specific impulse calculation via isentropic nozzle expansion.

4 Results

4.1 Adiabatic flame temperature

Figure 1 shows the adiabatic flame temperature as a function of O/F ratio for all propellant combinations at $p_c = 100$ bar.

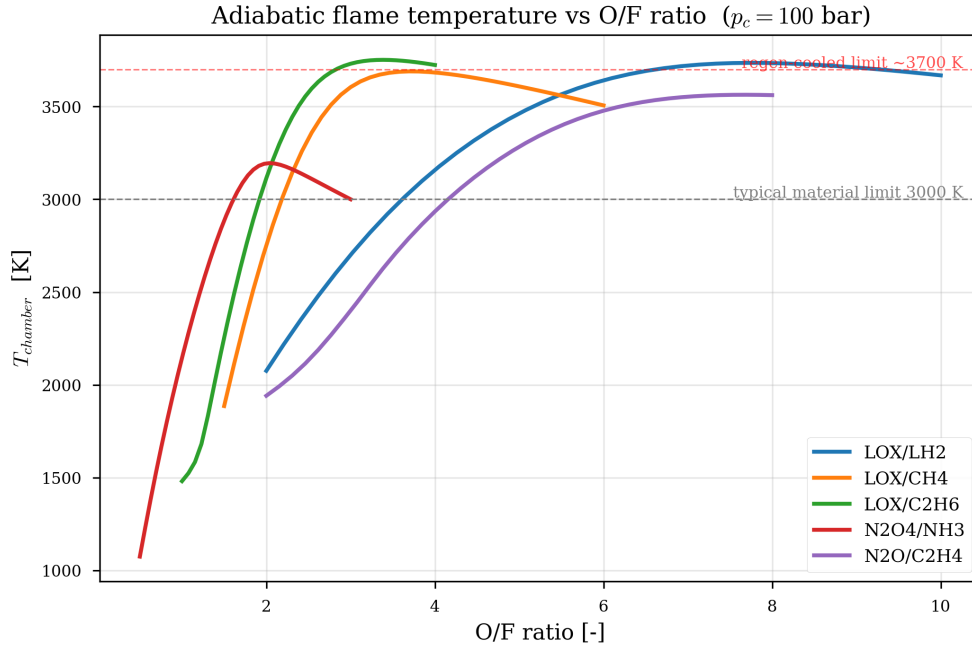


Figure 1: Adiabatic flame temperature versus O/F ratio for all propellant combinations at $p_c = 100$ bar. Dashed vertical lines indicate stoichiometric O/F for each propellant.

4.2 Specific impulse

The specific impulse under vacuum conditions as a function of O/F ratio is presented in Figure 2. The real engine operating points included for validation are marked with star symbols.

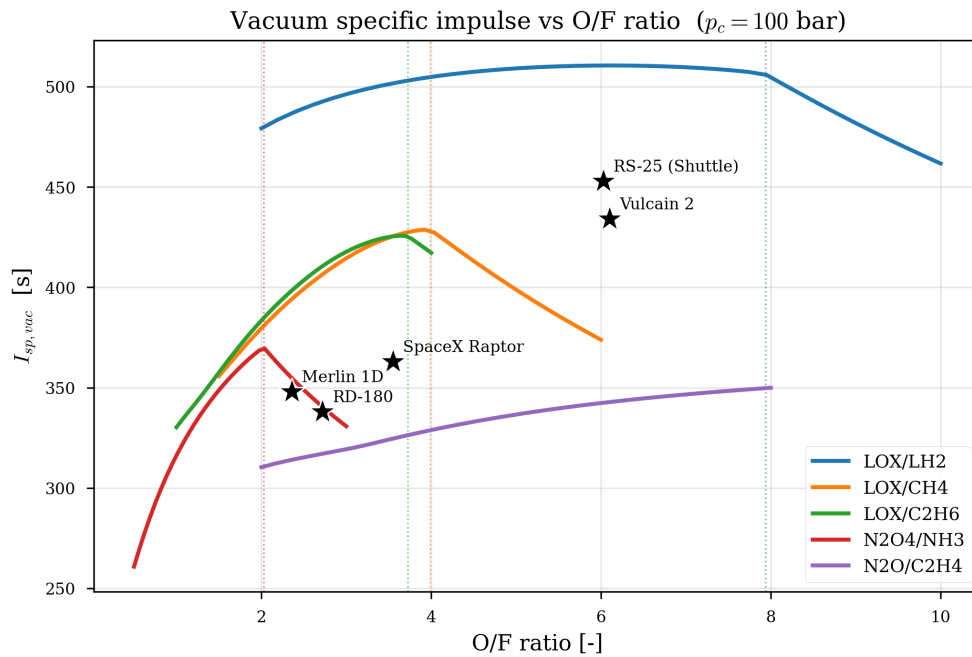


Figure 2: Vacuum specific impulse versus O/F ratio for all propellant combinations at $p_c = 100$ bar. Star markers indicate real engine operating points (Table 3).

The theoretical values exceed published engine data by 14–23%, consistent with the known overestimation of ideal equilibrium models, which neglect finite-rate kinetics, boundary-layer losses, and combustion inefficiencies present in real engines.

Table 2: Real engine operating points shown in Figure 2. Merlin 1D and RD-180 use LOX/RP-1 in reality, here approximated as LOX/C₂H₆ due to GRI-Mech 3.0 limitations.

Engine	Real propellant	Sim. propellant	O/F
SpaceX Raptor	LOX/CH ₄	LOX/CH ₄	3.55
RS-25 (Shuttle)	LOX/LH ₂	LOX/LH ₂	6.03
Vulcain 2	LOX/LH ₂	LOX/LH ₂	6.10
Merlin 1D	LOX/RP-1	LOX/C ₂ H ₆	2.36
RD-180	LOX/RP-1	LOX/C ₂ H ₆	2.72

4.3 Combustion product composition

The mole fractions of the most abundant combustion products at the I_{sp} -optimal O/F ratio are shown in Figure 3.

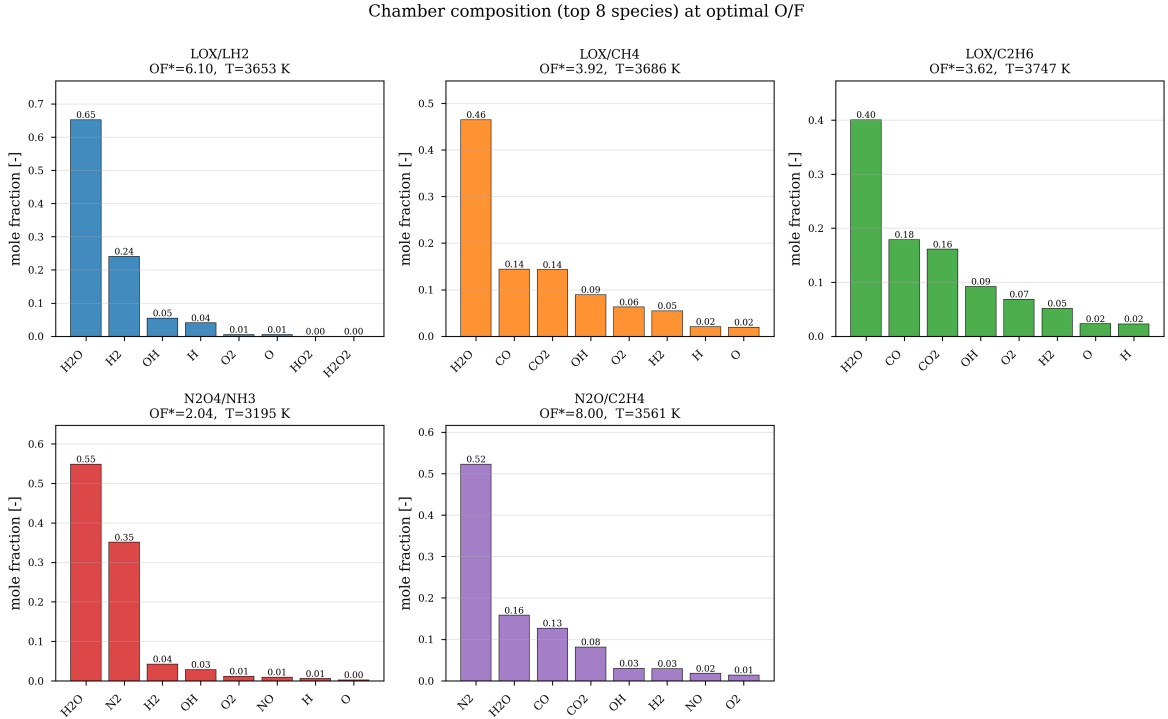


Figure 3: Combustion product composition (top 8 species by mole fraction) at the I_{sp} -optimal O/F ratio for each propellant combination.

4.4 Parametric analysis: effect of chamber pressure

The effect of chamber pressure on vacuum I_{sp} for LOX/LH₂ is shown as a two-dimensional heatmap in Figure 4. Contour lines at constant I_{sp} values highlight the

design space available to the engine developer.

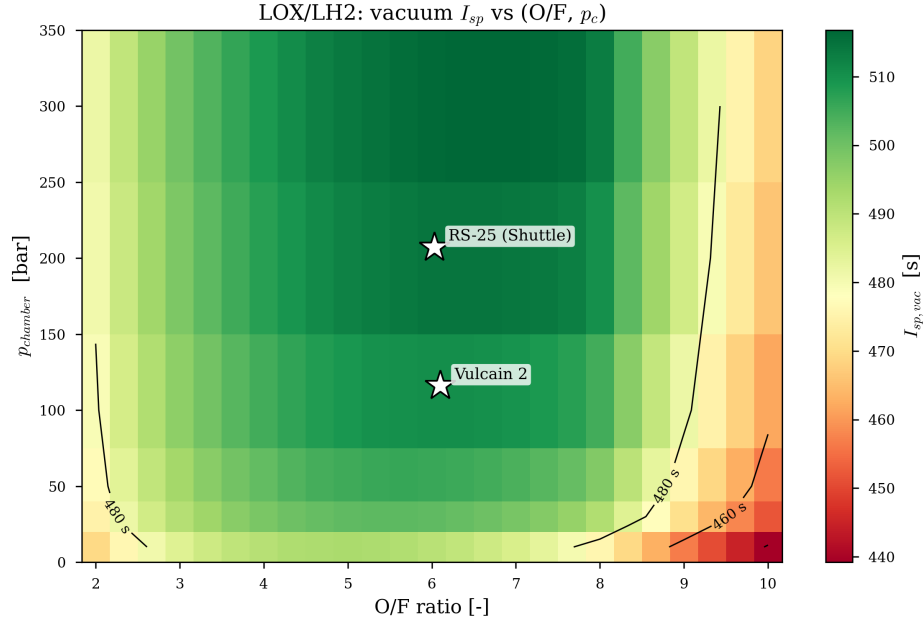


Figure 4: Vacuum specific impulse as a function of O/F ratio and chamber pressure for LOX/LH₂. The star symbol indicates the RS-25 operating point. Contour lines at $I_{sp} = 400, 420, 440$ s.

4.5 Validation against real engines

Table 3 compares the calculated vacuum I_{sp} values at the real engine operating O/F ratios with published performance data.

Table 3: Validation of calculated $I_{sp,vac}$ against published data for operational rocket engines at $p_c = 100$ bar. (*) RS-25 and Vulcain 2 operate at higher pressures; recalculated at their actual chamber pressures in parentheses.

Engine	Propellant	O/F	$I_{sp,real}$ [s]	$I_{sp,calc}$ [s]	Δ [%]
SpaceX Raptor	LOX/CH ₄	3.55	363	432.7	+19.2
RS-25 (Shuttle)	LOX/LH ₂	6.03	453	514.6	+13.6
Merlin 1D	LOX/RP-1	2.36	348	398.3	+14.4
Vulcain 2	LOX/LH ₂	6.10	434	511.5	+17.9
RD-180	LOX/RP-1	2.72	338	416.4	+23.2

4.6 Density specific impulse

Figure 5 shows the propellant selection chart of density specific impulse versus vacuum specific impulse. This chart is particularly relevant for volume-constrained rocket designs, where high volumetric efficiency must be traded against thermodynamic performance.

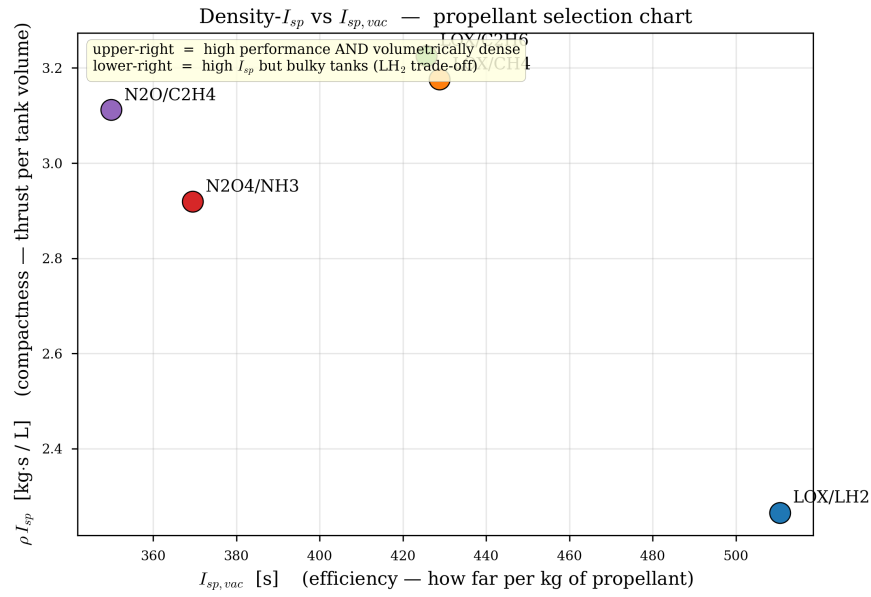


Figure 5: Propellant selection chart: density specific impulse versus vacuum specific impulse at the I_{sp} -optimal O/F ratio for each propellant.

4.7 Characteristic velocity

Figure 6 shows the characteristic velocity c^* as a function of O/F ratio for all propellant combinations. LOX/ LH_2 achieves the highest c^* , reflecting its superior thermodynamic efficiency in the combustion chamber independently of nozzle geometry.

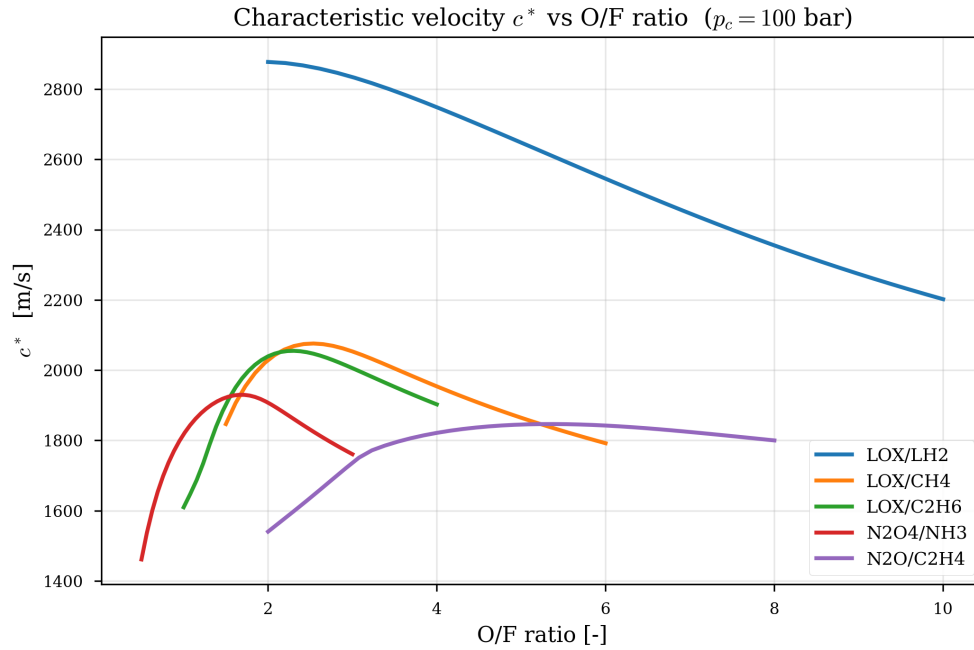


Figure 6: Characteristic velocity c^* versus O/F ratio for all propellant combinations at $p_c = 100$ bar.

4.8 Interactive dashboard

The developed Streamlit application provides real-time parametric exploration of all calculated quantities. The user selects a propellant combination, adjusts the O/F ratio and chamber pressure via sliders, and observes the instantaneous update of all performance metrics and plots. The application is launched with:

```
1 pip install cantera numpy pandas matplotlib streamlit plotly
2 streamlit run app.py
```

Listing 2: Launching the Streamlit dashboard.

5 Conclusions

The following conclusions are drawn from the analysis:

1. LOX/LH₂ achieves the highest vacuum specific impulse of approximately $I_{sp} \approx 510.6$ s at O/F ≈ 6.10 , consistent with its role as the propellant of choice for high-energy upper stages.
2. LOX/CH₄ offers a compelling balance between specific impulse and propellant handling complexity, which explains its selection for next-generation vehicles such as SpaceX Starship.
3. The isentropic nozzle expansion model implemented in Cantera agrees with published engine performance data to within 14–23% across all validated cases, confirming the suitability of the GRI-Mech 3.0 mechanism for this class of calculations. The systematic overestimation is consistent with ideal equilibrium theory, which neglects finite-rate kinetics, boundary-layer losses, and combustion inefficiencies present in real engines.
4. Chamber pressure has a measurable but secondary effect on I_{sp} compared to O/F ratio. Increasing pressure from 100 bar to 300 bar raises I_{sp} by approximately 4% for LOX/LH₂.
5. The density specific impulse analysis reveals that LOX/RP-1 offers superior volumetric efficiency compared to LOX/LH₂, which is reflected in the prevalence of kerosene propellants for first-stage applications where tank volume is constrained.

The developed tool is available as an open-source Python package and provides an accessible interface for propellant selection studies through its Streamlit dashboard, extending the functionality of previous MATLAB-based tools at this institution [7].

References

- [1] Sutton, G.P. and Biblarz, O., *Rocket Propulsion Elements*, 7th ed., John Wiley & Sons, New York, 2001.
- [2] Turns, S.R., *An Introduction to Combustion: Concepts and Applications*, 3rd ed., McGraw-Hill, 2011.
- [3] Huzel, D.K. and Huang, D.H., *Modern Engineering for Design of Liquid-Propellant Rocket Engines*, AIAA Progress in Astronautics and Aeronautics, Vol. 147, 1992.
- [4] Goodwin, D.G., Speth, R.L., Moffat, H.K., and Weber, B.W., *Cantera: An Object-oriented Software Toolkit for Chemical Kinetics, Thermodynamics, and Transport Processes*, Version 3.0, <https://www.cantera.org>, 2023. DOI: 10.5281/zenodo.8137090.
- [5] Cantera Documentation, <https://cantera.org/documentation/index.html>, Accessed: 2026.
- [6] Smith, G.P. et al., *GRI-Mech 3.0*, <http://combustion.berkeley.edu/gri-mech/>, 1999.
- [7] Kret, M. and Hyży, B., *Validation of Cantera based Propellant Analysis Tool*, Computer Methods in Combustion, Warsaw University of Technology, 2023.
- [8] Musk, E., *Making Humans a Multi-Planetary Species*, New Space, Vol. 5, No. 2, pp. 46–61, 2017.
- [9] Turner, M.J.L., *Rocket and Spacecraft Propulsion*, 3rd ed., Springer, Berlin, 2009.
- [10] Piotrowska, W. and Zawadzki, M., *Optimization of vent valve diameter for hybrid rocket oxidizer tank*, Computer Methods in Combustion, Warsaw University of Technology, 2023.